

Deep Residual SRCNN for Single-Image Super-Resolution

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Abstract

We reproduce the classical Super-Resolution Convolutional Neural Network (SRCNN) and evaluate controlled extensions in depth and residual learning. Images are represented at 224x224, with 112x112 bilinear degradation followed by upsampling. On the pilot 20-image test split, SRCNN reaches 25.26 dB, stabilized Deep SRCNN 25.45 dB, and Deep Residual SRCNN 27.15 dB. A final 800-image training run reaches 27.75 dB (MSE 0.001680) on a fixed 100-image evaluation collection. Re-evaluating the 60-image residual model on the same collection gives 26.99 dB, so the larger training set adds 0.76 dB and reduces MSE by about 16%.

1 Introduction

Single-image super-resolution (SISR) reconstructs a high-resolution image from a degraded observation. Interpolation increases spatial resolution but cannot recover detail lost during degradation. SRCNN instead learns a direct mapping from an interpolated low-resolution image to a high-resolution target using convolutional feature extraction, nonlinear mapping, and reconstruction. This study reproduces the classical SRCNN pipeline and tests two controlled extensions: deeper convolutional processing and a global residual connection. The goal is not to claim state-of-the-art performance, but to isolate how architectural changes and training-set size affect a small, reproducible SISR system.

2 Method

Preprocessing. Images are converted to RGB, center-cropped to 800x800, resized to 224x224, downsampled to 112x112 with bilinear interpolation, then upsampled back to 224x224 before neural reconstruction. Targets are the corresponding 224x224 crops. The pilot split uses 60 training, 20 validation, and 20 test images from a DIV2K-derived collection; the final experiment uses 800 training, 100 validation, and a fixed 100-image evaluation collection.



Figure 1. Common image pipeline: HR target, degraded 112x112 image, and 224x224 interpolated input.

Optimization. All neural models minimize pixel-level mean-squared error (MSE) with Adam, learning rate $3e-4$, gradient clipping at 1.0, and 50 epochs. For normalized images, PSNR = $10 \log_{10}(1/\text{MSE})$.

3 Model Variants

SRCNN. Three convolutions: 3→64 with a 9x9 kernel, 64→32 with a 1x1 kernel, and 32→3 with a 5x5 kernel; ReLU follows the first two layers. **Deep SRCNN** inserts two 64-channel 3x3 convolutions. **Deep Residual SRCNN** keeps the five-layer backbone and predicts a correction, $\text{SR} = X + F(X)$, allowing the network to focus on missing high-frequency content while retaining coarse structure supplied by interpolation.



Figure 2. Pilot qualitative comparison across interpolation baselines and SRCNN variants.

4 Training Behavior

The deeper network provides a modest gain over the baseline, while the residual formulation produces the largest architectural improvement. Validation curves remain broadly stable under the clipped Adam training schedule.

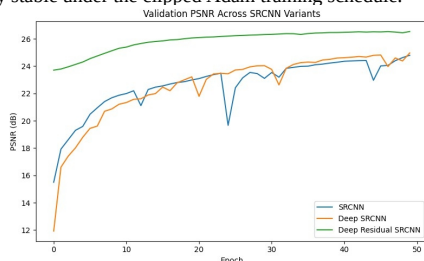


Figure 3. Validation PSNR across the three neural architectures on the pilot run.

5 Quantitative Results

On the original 20-image held-out pilot split, the classical SRCNN reaches 25.26 dB PSNR, stabilized Deep SRCNN reaches 25.45 dB, and Deep Residual SRCNN reaches 27.15 dB. Bicubic interpolation reaches 25.25 dB, providing a strong non-neural baseline.

Model	MSE	PSNR (dB)
Nearest Neighbor	0.003608	24.43
Bilinear	0.003663	24.36
Bicubic	0.002985	25.25
SRCNN	0.002975	25.26
Deep SRCNN	0.002850	25.45
Deep Residual SRCNN - 60*	0.002000	26.99
Deep Residual SRCNN - 800	0.001680	27.75

* The 60-image residual model is re-evaluated here on the same fixed 100-image collection used by the 800-image model; its pilot 20-image score is 27.15 dB.

6 Data-Scale Effect

The final Deep Residual SRCNN is trained on 800 images while holding the evaluation collection fixed. On that common test set, the 60-image model obtains 26.99 dB and the 800-image model obtains 27.75 dB: a +0.76 dB improvement and approximately 16% lower MSE.

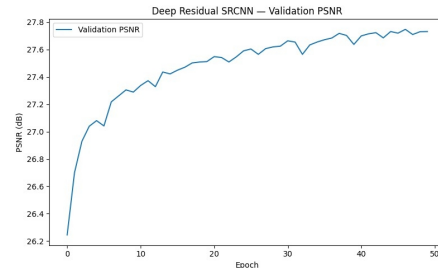


Figure 4. Validation PSNR for the 800-image Deep Residual SRCNN run.



Figure 5. Qualitative comparison: SRCNN, Deep SRCNN, Deep Residual SRCNN, and ground truth.

7 Discussion and Conclusion

The experiments separate three effects. First, replacing hand-designed interpolation with a learned SRCNN mapping gives a small gain over bicubic on the pilot split. Second, adding convolutional depth alone gives only a modest improvement. Third, global residual learning produces the strongest architectural gain, consistent with the model learning a correction rather than reconstructing the entire image from scratch. Finally, increasing the training set from 60 to 800 images improves performance on the same evaluation collection. Together, these findings support a simple conclusion: within this controlled SRCNN pipeline, residual learning matters substantially, and data scale provides an additional measurable benefit.

References

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